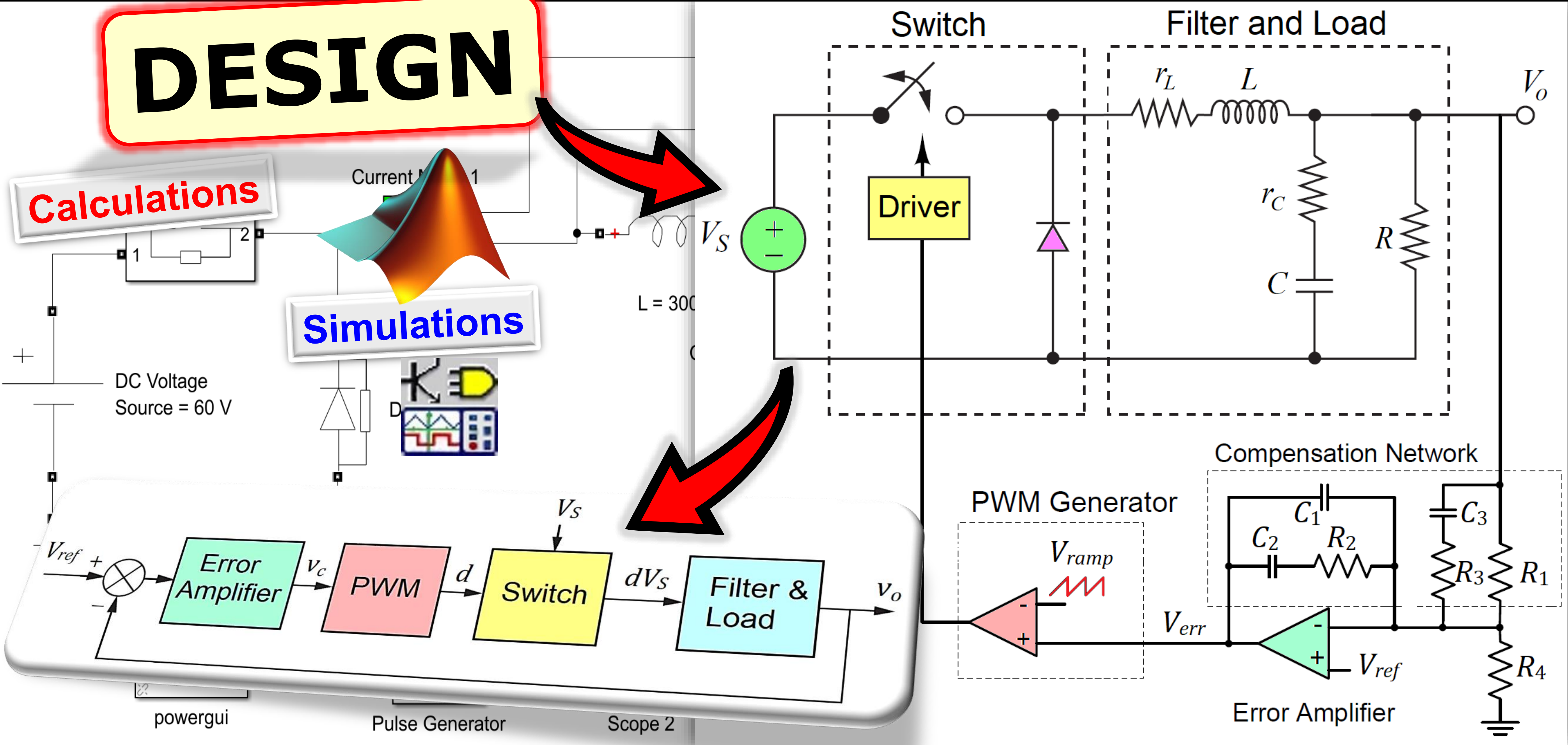
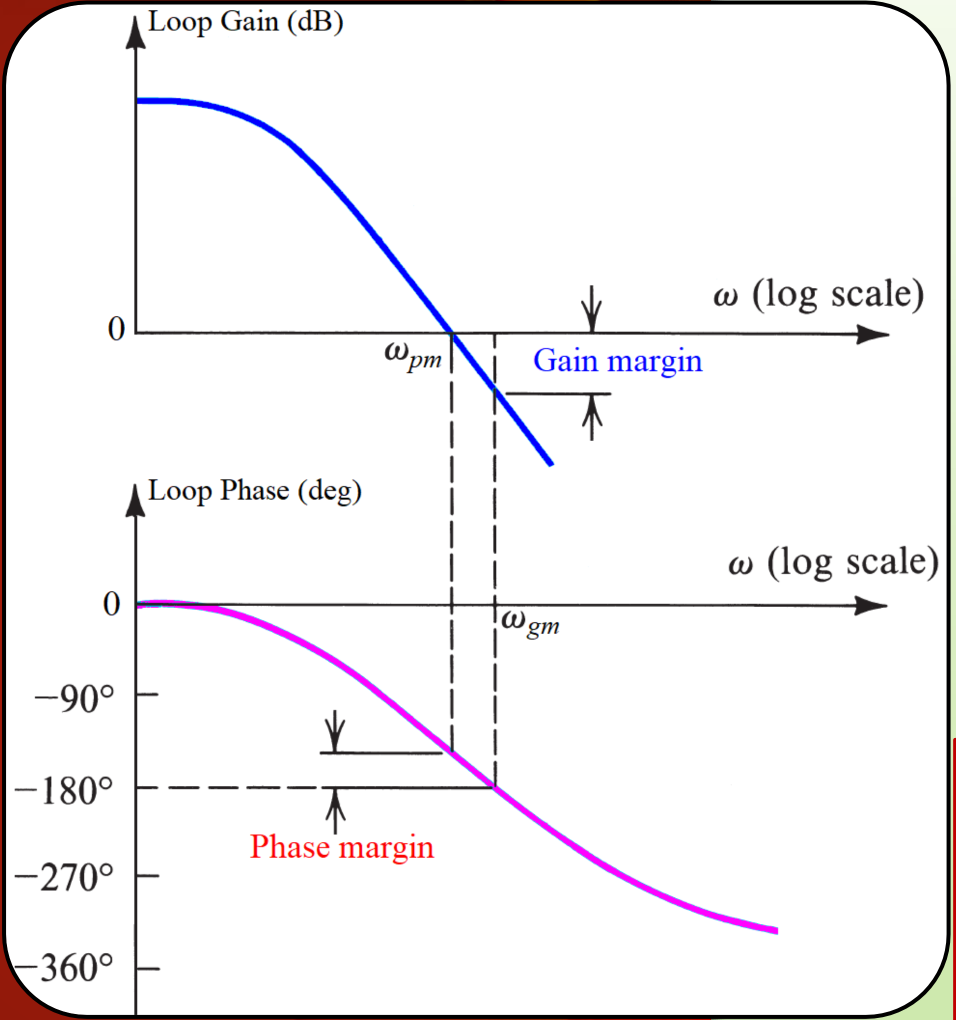
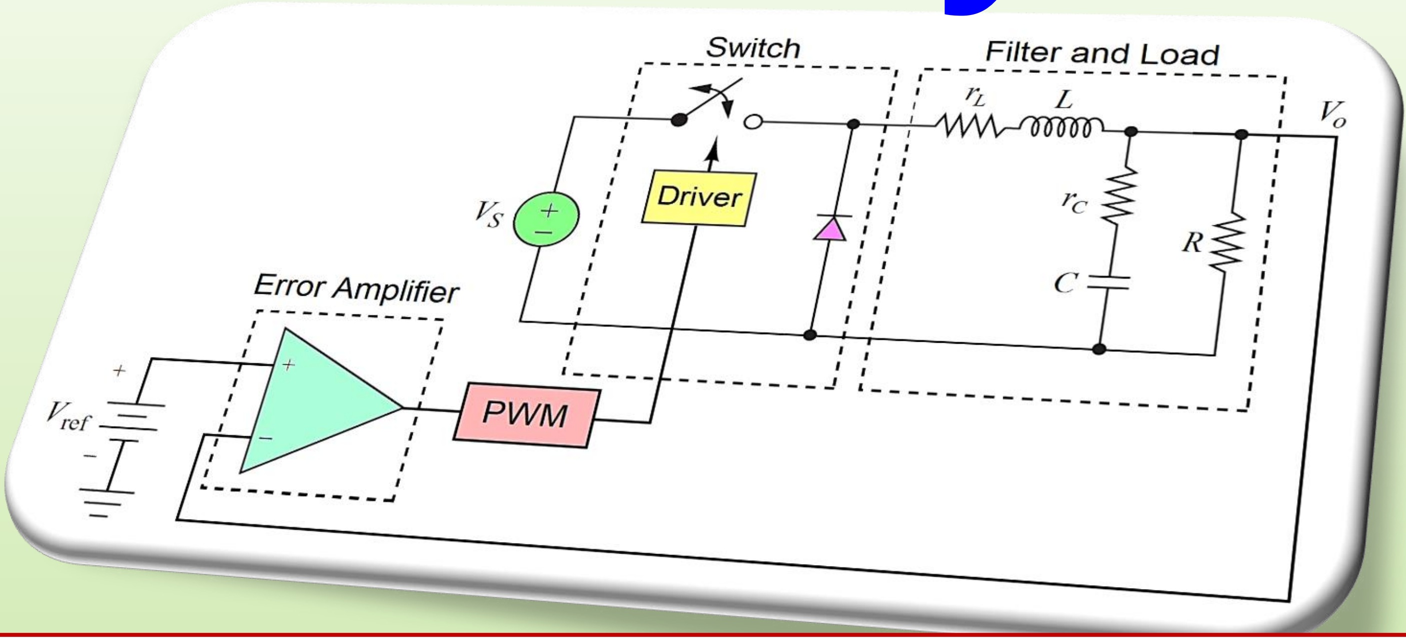
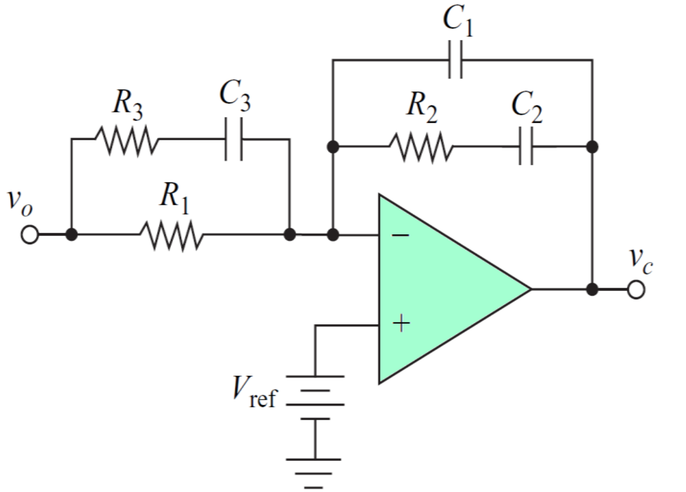
TYPE 3 COMPENSATOR



Theory

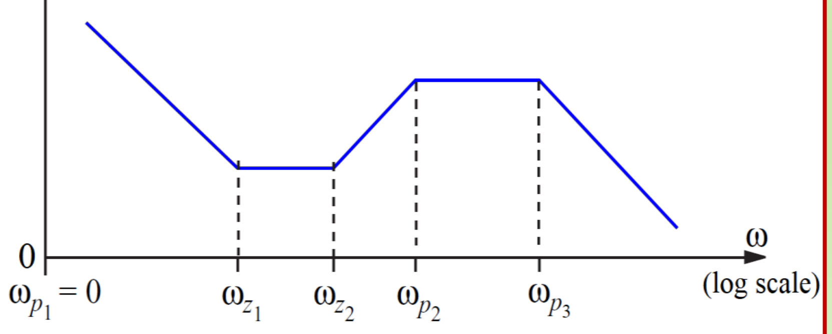






$\left|\frac{v_c}{v_o}\right|$

$$G_3(s) = -\frac{(R_1 + R_3)(s + \omega_{z1})(s + \omega_{z2})}{R_1 R_3 C_1 s(s + \omega_{p2})(s + \omega_{p3})}$$

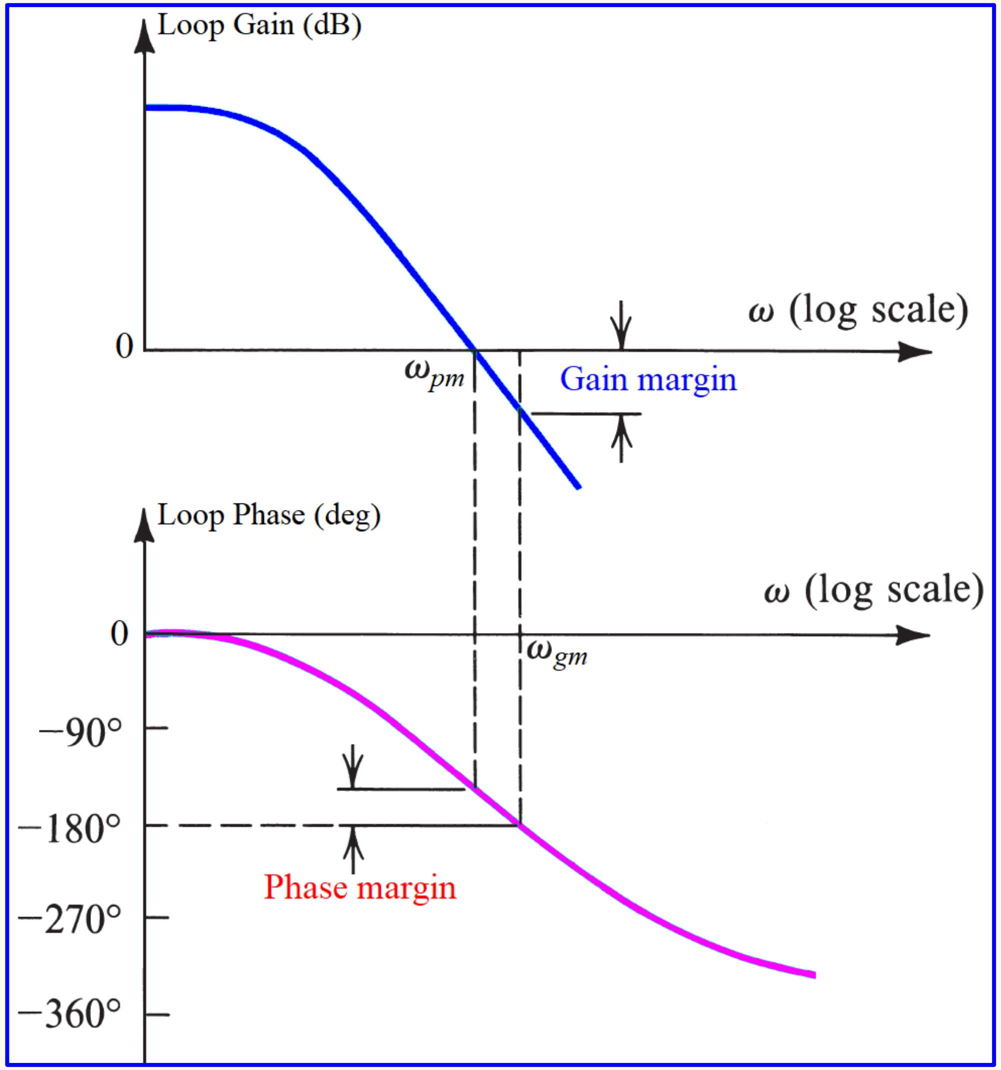


1. Frequency Response Analysis

Stability Margins

- **Gain Margin:**
  1. Find the **frequency** where the **loop phase** is $-180°$. This frequency is called ω_{gm}.
  2. Determine the **loop gain** at ω_{gm}.
  3. The **gain margin (GM)** is the gain required to ***raise*** the **magnitude curve** to 0 dB.

- **Phase Margin:**
  1. Find the **frequency** where the **loop gain** is 0 dB. This frequency is called ω_{pm}.
  2. The **phase margin (PM)** is the **difference** between the **loop phase** at ω_{pm} and $-180°$.



Example:

Loop transfer function: $L(j\omega) = \dfrac{1000}{(j\omega+4)(j\omega+8)(j\omega+10)}$

Determine:
- Phase margin (*PM*)
- Gain margin (*GM*)

Phase Margin: $|L(j\omega)| = 1 \Rightarrow \dfrac{1000}{\sqrt{\omega^2+4^2}\sqrt{\omega^2+8^2}\sqrt{\omega^2+10^2}} = 1$

Solving:
$\omega = 6.79 \text{ rad/s} \equiv \omega_{pm}$

$$\arg\left(L(j\omega_{pm})\right) = -\arctan\left(\frac{\omega_{pm}}{4}\right) - \arctan\left(\frac{\omega_{pm}}{8}\right) - \arctan\left(\frac{\omega_{pm}}{10}\right) = -134°$$

$$PM = 180 + \arg\left(L(j\omega_{pm})\right) = 46°$$

Gain Margin: $\arg(L(j\omega)) = -\arctan\left(\frac{\omega}{4}\right) - \arctan\left(\frac{\omega}{8}\right) - \arctan\left(\frac{\omega}{10}\right) = -180°$

Solving:
$\omega = 12.33 \text{ rad/s} \equiv \omega_{gm}$

$$|L(j\omega_{gm})| = \frac{1000}{\sqrt{\omega_{gm}^2+4^2}\sqrt{\omega_{gm}^2+8^2}\sqrt{\omega_{gm}^2+10^2}} = 0.3307\ (-9.61 \text{ dB})$$

$$GM = \frac{1}{|L(j\omega_{gm})|} = 3.024\ (9.61 \text{ dB})$$

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1. Frequency Response Analysis

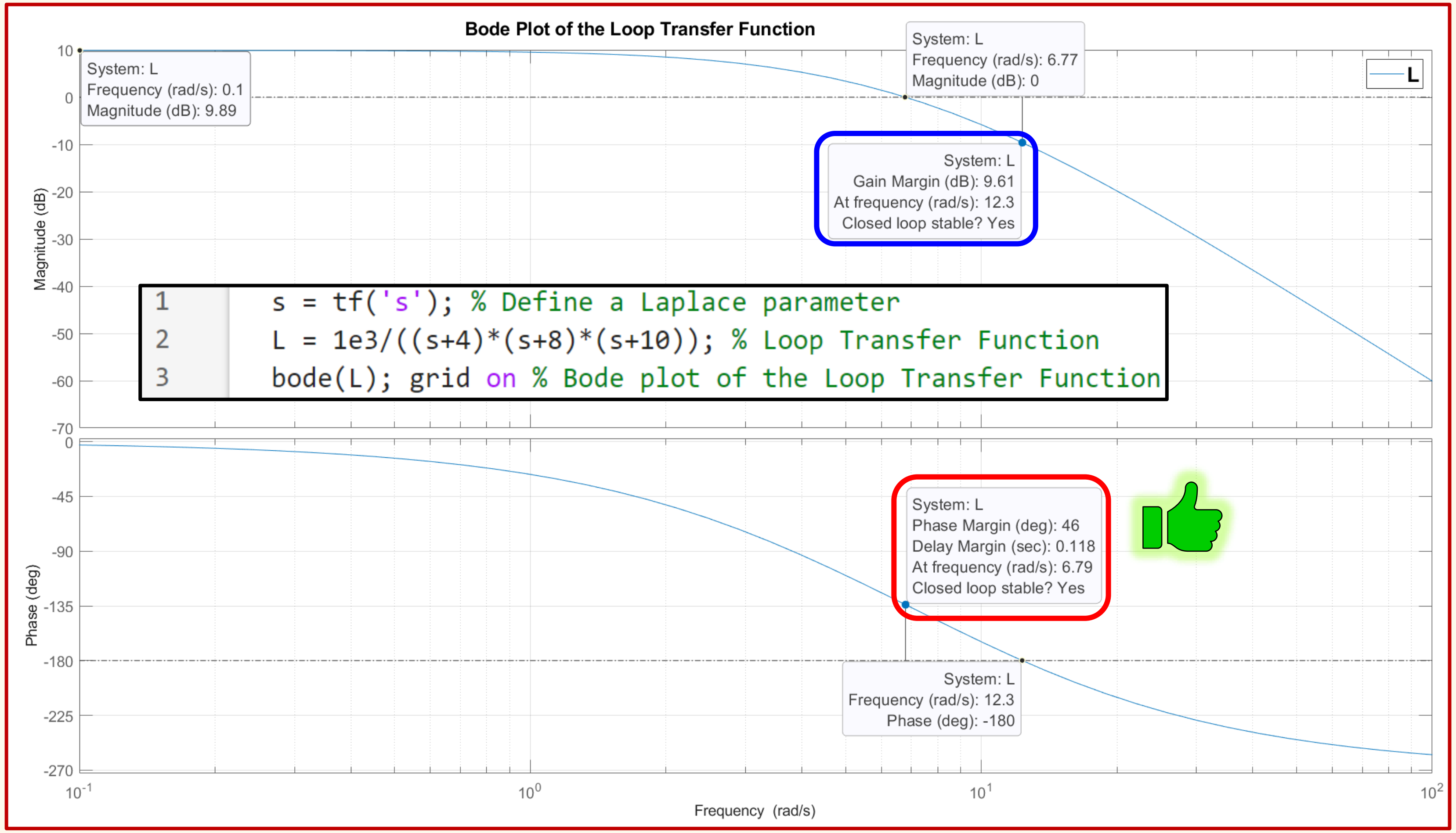
Example:

Loop transfer function: $L(j\omega) = \dfrac{1000}{(j\omega+4)(j\omega+8)(j\omega+10)}$

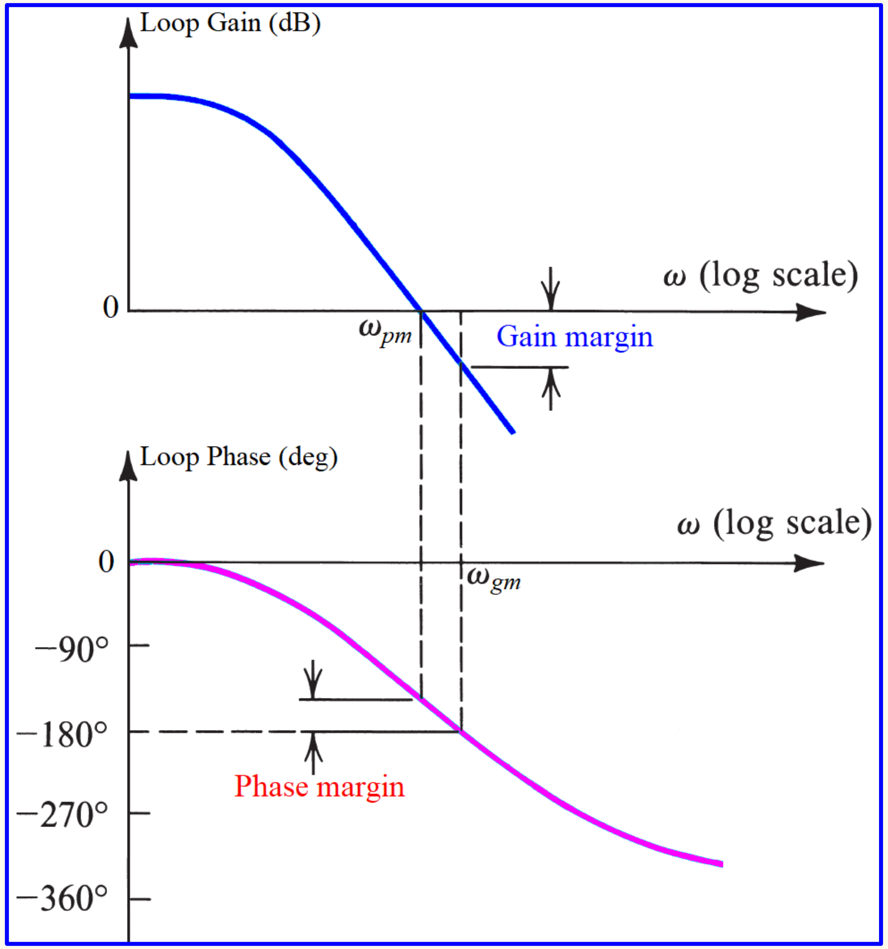
Determine:

- Phase margin (*PM*)
- Gain margin (*GM*)

Simulation Results:




```
s = tf('s'); % Define a Laplace parameter
L = 1e3/((s+4)*(s+8)*(s+10)); % Loop Transfer Function
bode(L); grid on % Bode plot of the Loop Transfer Function
```

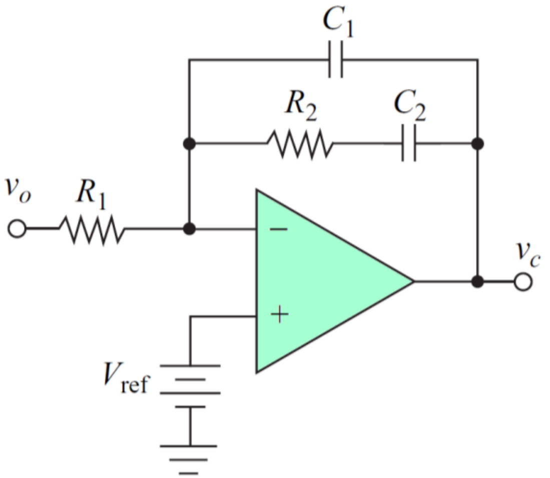


$\omega_{pm} = 6.79\ \text{rad/s}$ $\quad PM = 180 + \arg\left(L(j\omega_{pm})\right) = 46°$

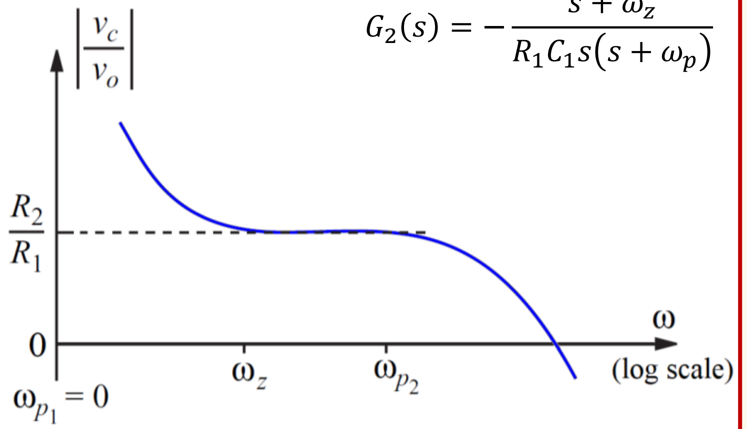
$\omega_{gm} = 12.33\ \text{rad/s}$ $\quad GM = \dfrac{1}{|L(j\omega_{gm})|} = 3.024\ (9.61\ \text{dB})$

2. Controller Circuits

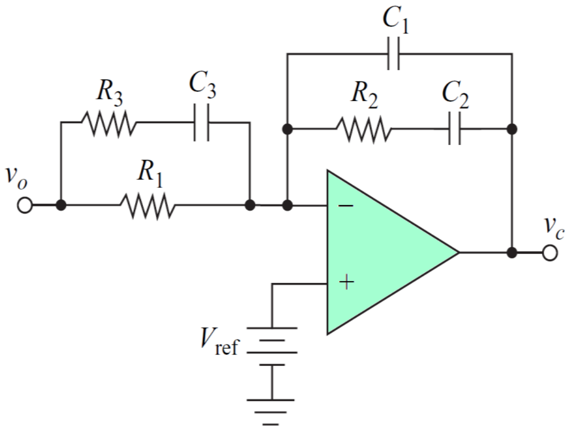
❖ Type 2 Compensated Error Amplifier



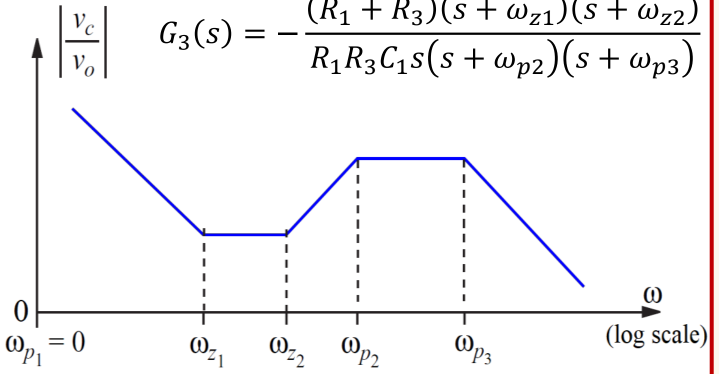
$$G_2(s) = -\frac{s+\omega_z}{R_1 C_1 s(s+\omega_p)}$$

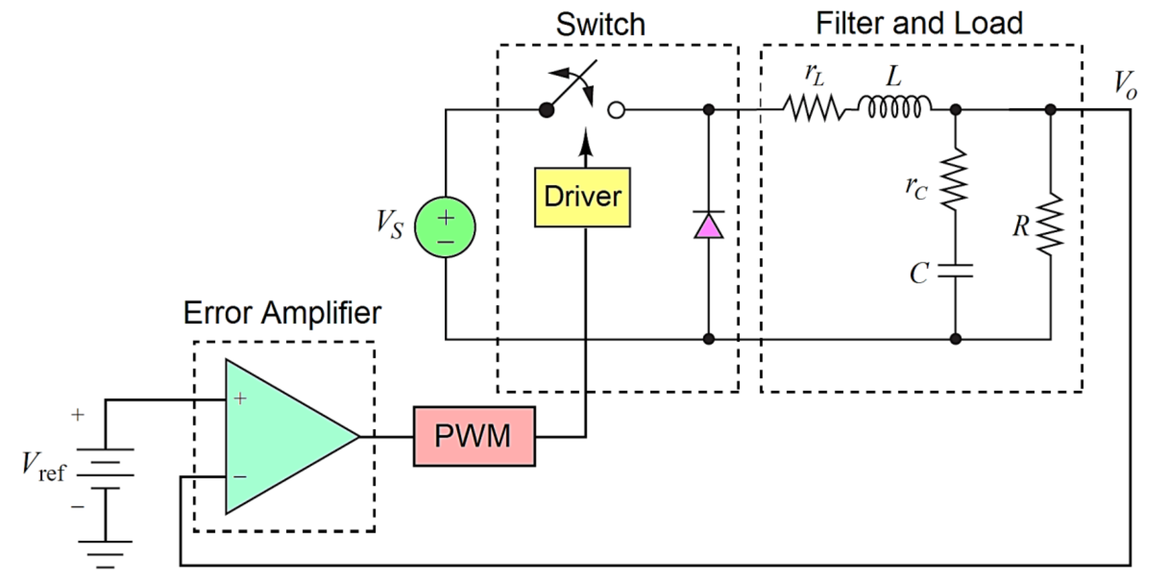


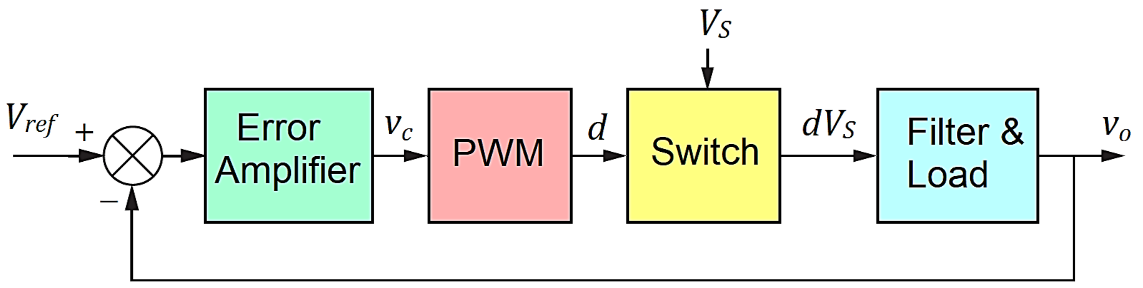
❖ Type 3 Compensated Error Amplifier



$$G_3(s) = -\frac{(R_1+R_3)(s+\omega_{z1})(s+\omega_{z2})}{R_1 R_3 C_1 s(s+\omega_{p2})(s+\omega_{p3})}$$

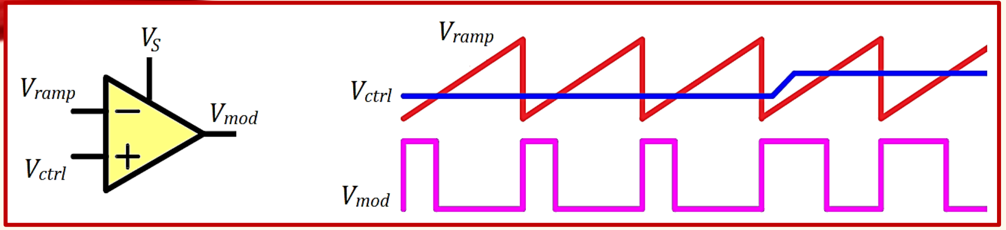






2. Controller Circuits

❖ **Modulator:**



Modulator Average Output Voltage: $V_{mod} = \frac{1}{T}\int_0^T v_{mod}(t)dt = \frac{1}{T}\int_0^{t_{on}} V_S dt$

Modulator Transfer Function: $M(s) = \frac{\Delta V_{mod}}{\Delta V_{ctrl}}$

$$M(s) = \frac{dV_{mod}}{dV_{ctrl}} = \frac{d}{dV_{ctrl}}\left(V_S \frac{V_{ctrl}}{V_{ramp}}\right) \qquad \boxed{M(s) = \frac{V_S}{V_{ramp}}}$$

❖ **Filter & Load:**

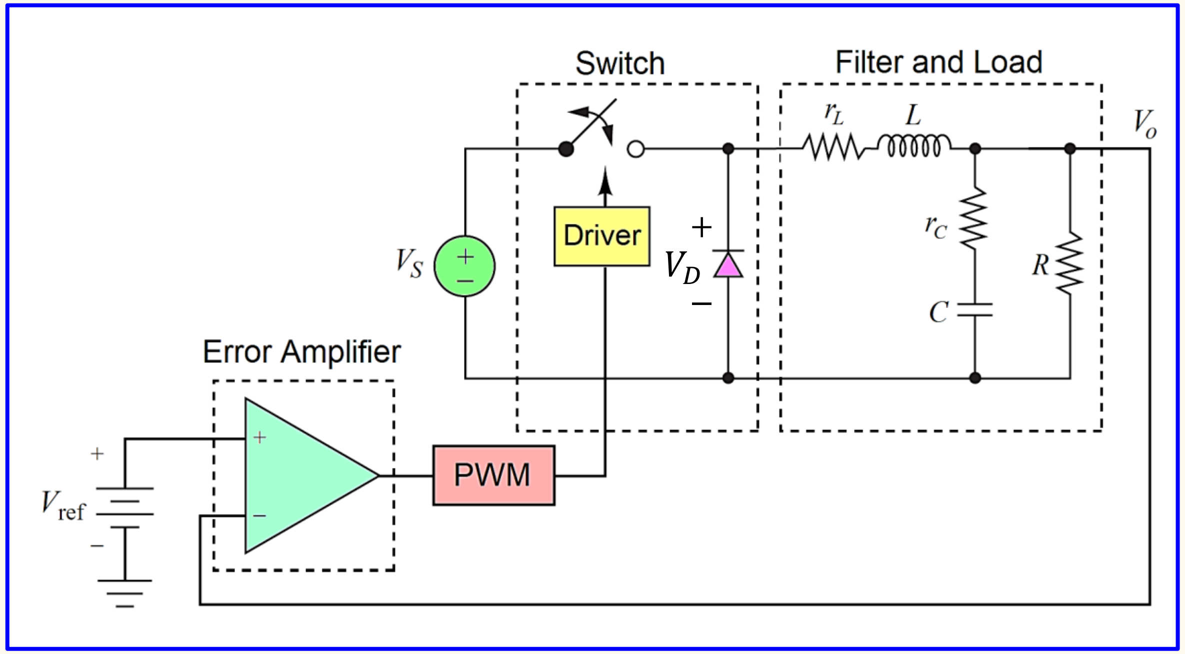
Filter & Load Transfer Function: $F(s) = \frac{V_o}{V_D} = \frac{1 + r_C C s}{1 + \frac{r_L}{R} + \left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)s + \left(1 + \frac{r_C}{R}\right)LCs^2}$

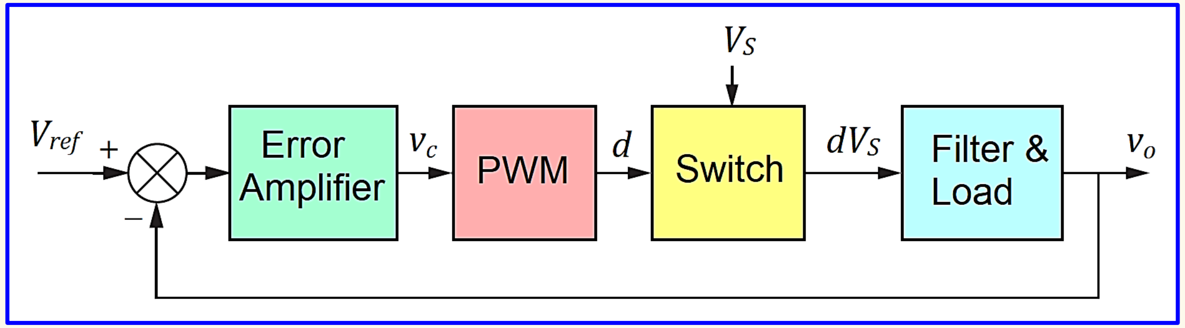
❖ **Compensator:**

Compensator Transfer Function: $G(s) = \frac{V_c}{V_o} = \frac{A_{OL}(s)}{1 + A_{OL}(s)\beta(s)} \approx \frac{A_{OL}(s)}{A_{OL}(s)\beta(s)} \approx \frac{1}{\beta(s)}$

Error Amplifier Open-Loop Transfer Function: $A_{OL}(s) = \frac{A_{DC}}{1 + \frac{s}{\omega_{pe}}}$ $\qquad A_{OL}(s)\beta(s) \gg 1$

Feedback Transfer Function: $\beta(s)$





Calculations

Given:

- DC input voltage $V_S = 60\text{ V}$
- Average output voltage $V_o = 15\text{ V}$
- Average output current $I_o = 2\text{ A}$
- Maximum percentage peak-peak output ripple voltage $\Delta V_o / V_o = 0.01\ (1\%)$

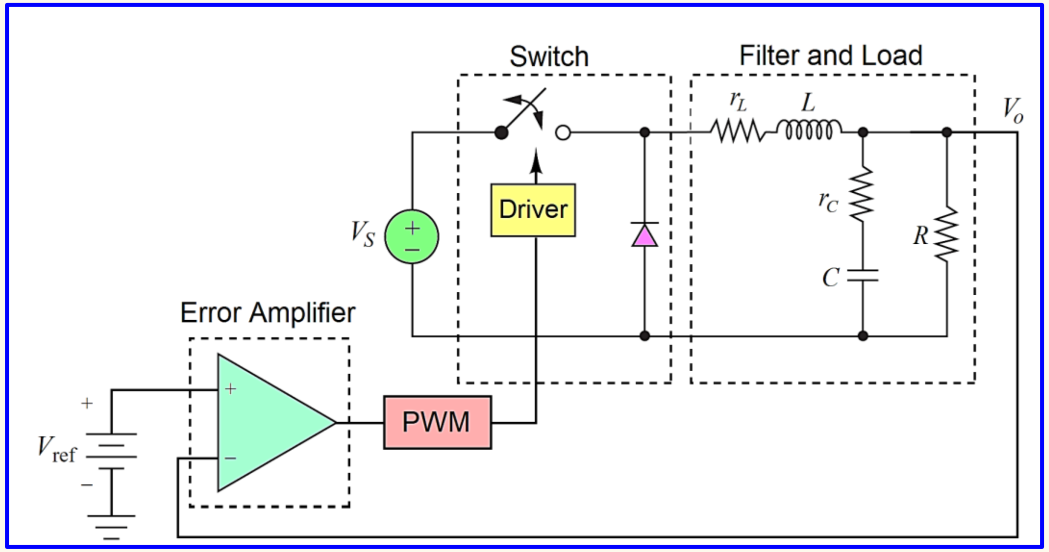
Buck Converter Component Values & Selections:

$L = 300\ \mu\text{H}$ $\quad r_L = 25\text{ m}\Omega$ $\quad R = V_o/I_o = 7.5\ \Omega$

$C = 20\ \mu\text{F}$ $\quad r_C = 400\text{ m}\Omega$ $\quad f_s = 100\text{ kHz}$ (selected)

Design Objective:

- **Type 3 Compensated Error Amplifier** for a **stable control system**.
- Design for a **crossover frequency** of 10 kHz and a **phase margin** of $55°$



Solution: ☐ Calculations

Filter & Load Transfer Function:

$$F(s) = \frac{1 + r_C C s}{1 + \frac{r_L}{R} + \left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)s + \left(1 + \frac{r_C}{R}\right)LCs^2}$$

$$F(j\omega) = \frac{1 + j\omega r_C C}{1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right)LC\omega^2 + j\omega\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)}$$

Modulator Transfer Function:

$$M(s) = \frac{V_S}{V_{ramp}}$$

Type 3 Compensated Error Amplifier Transfer Function:

$$G_3(s) = -\frac{R_1 + R_3}{R_1 R_3 C_1} \cdot \frac{\left(s + \frac{1}{(R_1 + R_3)C_3}\right)\left(s + \frac{1}{R_2 C_2}\right)}{s\left(s + \frac{1}{R_3 C_3}\right)\left(s + \frac{C_1 + C_2}{C_1 C_2 R_2}\right)}$$

Assuming:
$R_1 \gg R_3$
$C_2 \gg C_1$

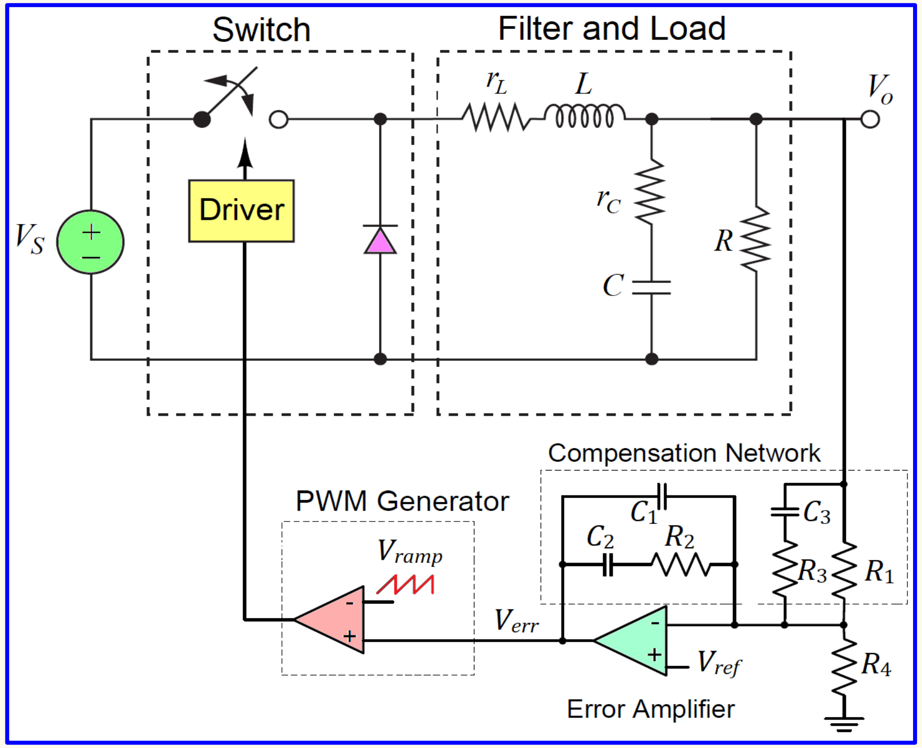
$$G_3(s) \approx -\frac{1}{R_3 C_1} \cdot \frac{\left(s + \frac{1}{R_1 C_3}\right)\left(s + \frac{1}{R_2 C_2}\right)}{s\left(s + \frac{1}{R_3 C_3}\right)\left(s + \frac{1}{R_2 C_1}\right)}$$

Loop Transfer Function:

$$L(s) = M(s)F(s)G_3(s)$$

$$L(s) = \frac{V_S}{V_{ramp}} \cdot \frac{1 + r_C C s}{1 + \frac{r_L}{R} + \left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)s + \left(1 + \frac{r_C}{R}\right)LCs^2} \cdot -\frac{1}{R_3 C_1} \cdot \frac{\left(s + \frac{1}{R_1 C_3}\right)\left(s + \frac{1}{R_2 C_2}\right)}{s\left(s + \frac{1}{R_3 C_3}\right)\left(s + \frac{1}{R_2 C_1}\right)}$$

$$L(j\omega) = \frac{V_S}{V_{ramp}} \cdot \frac{1 + j\omega r_C C}{1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right)LC\omega^2 + j\omega\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)} \cdot -\frac{1}{R_3 C_1} \cdot \frac{\left(j\omega + \frac{1}{R_1 C_3}\right)\left(j\omega + \frac{1}{R_2 C_2}\right)}{j\omega\left(j\omega + \frac{1}{R_3 C_3}\right)\left(j\omega + \frac{1}{R_2 C_1}\right)}$$



Given:
- DC input voltage $V_S = 60\ \text{V}$
- Average output voltage $V_o = 15\ \text{V}$
- Average output current $I_o = 2\ \text{A}$
- Maximum percentage peak-peak output ripple voltage $\Delta V_o / V_o = 0.01\ (1\%)$

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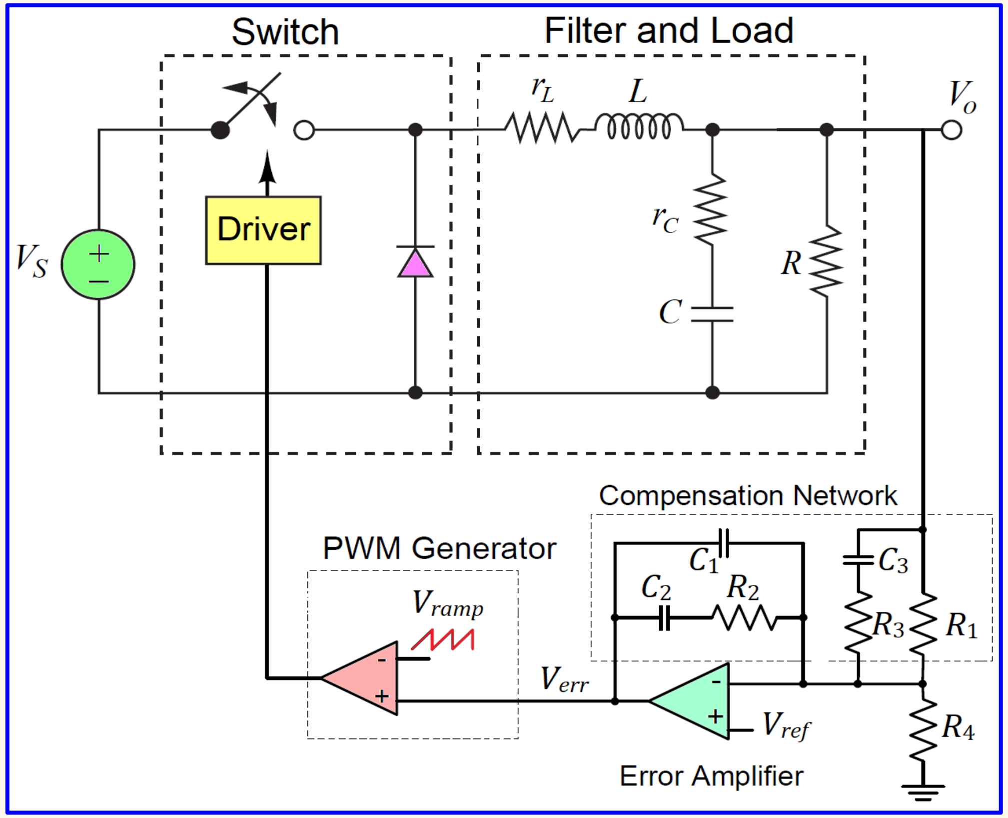
- **Type 3 Compensated Error Amplifier** for a **stable control system**.
- Design for a **crossover frequency** of $10\ \text{kHz}$ and a **phase margin** of $55°$



Solution: ❑ Calculations

Loop Transfer Function:

$$L(j\omega) = \frac{V_S}{V_{ramp}} \cdot \frac{1 + j\omega r_C C}{1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right) LC\omega^2 + j\omega\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)} \cdot -\frac{1}{R_3 C_1} \cdot \frac{\left(j\omega + \frac{1}{R_1 C_3}\right)\left(j\omega + \frac{1}{R_2 C_2}\right)}{j\omega\left(j\omega + \frac{1}{R_3 C_3}\right)\left(j\omega + \frac{1}{R_2 C_1}\right)}$$

Phase: $\arg(L(j\omega)) = \arg(M(j\omega)) + \arg(F(j\omega)) + \arg(G_3(j\omega))$

$$\arg(M(j\omega)) = 0°$$

$$\arg(F(j\omega)) = \arctan(\omega r_C C) - \left[\arctan\left(\frac{\omega\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)}{1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right) LC\omega^2}\right) + 180°\right]$$

Crossover Frequency
$f_t = 10\ \text{kHz}$

$$\arg(F(j\omega_t)) = \arctan(\omega_t r_C C) - \left[\arctan\left(\frac{\omega_t\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)}{1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right) LC\omega_t^2}\right) + 180°\right] = -146°$$

$$PM = 180° + \arg(L(j\omega_t)) \Rightarrow \arg(L(j\omega_t)) = PM - 180° = 55° - 180° = -125°$$

$$-125° = 0° - 146° + \arg(G_3(j\omega)) \Rightarrow \arg(G_3(j\omega)) = 21°$$

$$\phi_{comp} = 21° + 180° = 201°$$

Thus, the compensator needs to provide 201°.

Magnitude: $|L(j\omega)| = |M(j\omega)| \cdot |F(j\omega)| \cdot |G_3(j\omega)|$

The **loop gain** must be **unity** at the **crossover frequency**.

$$|F(j\omega)| = \frac{\sqrt{1 + (\omega r_C C)^2}}{\sqrt{\left(1 + \frac{r_L}{R} - \left(1 + \frac{r_C}{R}\right) LC\omega^2\right)^2 + \left(\omega\left(\frac{L}{R} + (r_L + r_C)C + \frac{r_L r_C C}{R}\right)\right)^2}}$$

$$|F(j\omega_t)| = 0.04636\ (-26.68\ \text{dB})$$

Modulator gain: $M = \frac{V_S}{V_{ramp}} = 15\ (23.52\ \text{dB})$ for LM5146

At the **crossover frequency**, the compensator needs to provide a gain of

$$|G_3(j\omega_t)| = \frac{1}{|M(j\omega_t)| \cdot |F(j\omega_t)|} = \frac{1}{15 \cdot 0.04636} = 1.438\ (3.155\ \text{dB})$$

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Given:

- DC input voltage $V_S = 60\ \text{V}$
- Average output voltage $V_o = 15\ \text{V}$
- Average output current $I_o = 2\ \text{A}$
- Maximum percentage peak-peak output ripple voltage $\Delta V_o / V_o = 0.01\ (1\%)$

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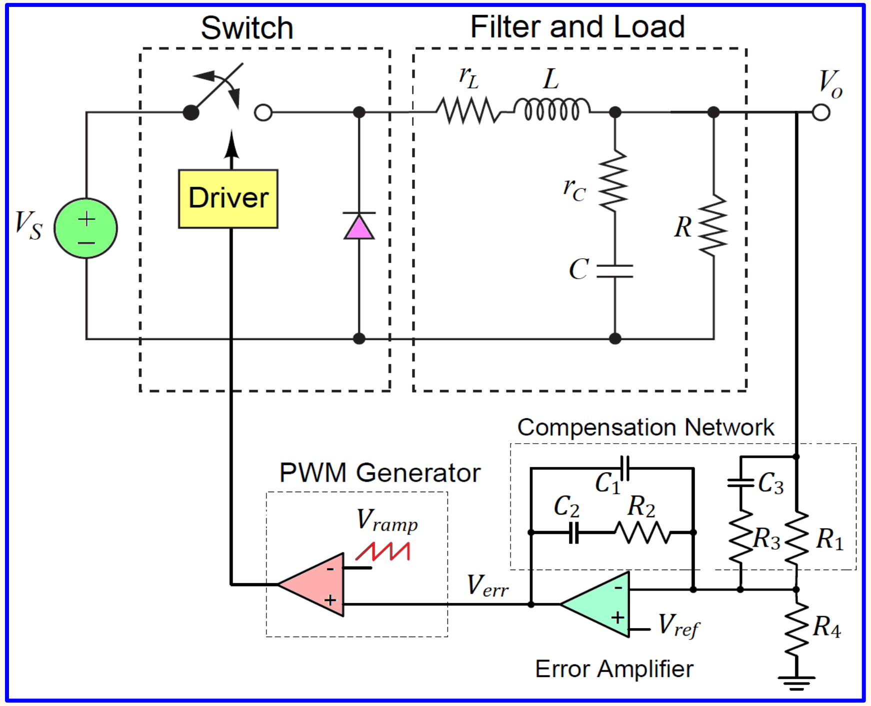
$\phi_{comp} = 201°$

$|G_3(j\omega_t)| = 1.438$

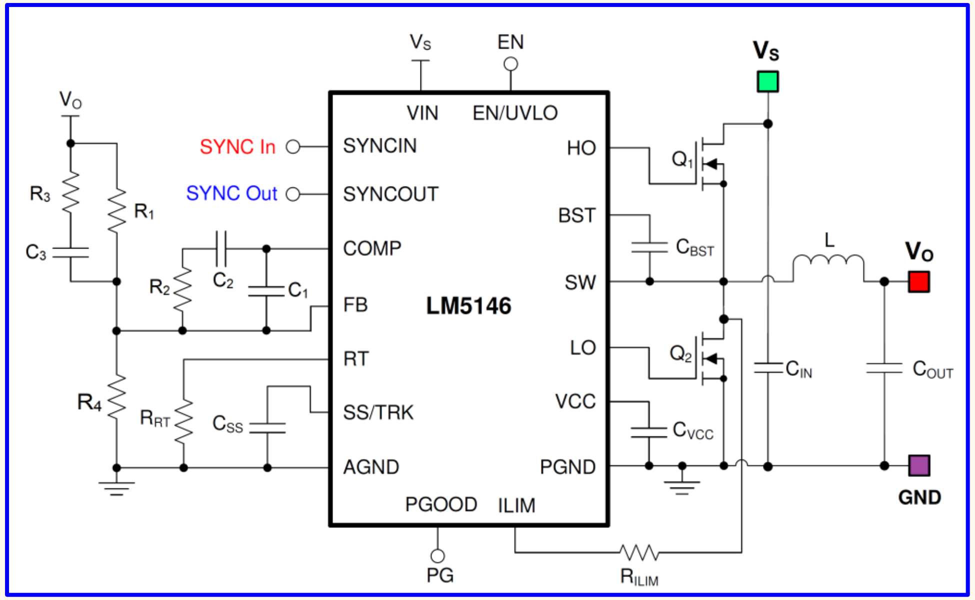


Solution: ❑ Calculations

Buck Controller IC (LM5146):

- Input Voltage Range: 5.5 V to 100 V
- Maximum Output Current: 20 A
- Control Mode: Voltage Mode
- Reference Voltage: $V_{ref} = 0.8\ \text{V}$
- PWM Ramp Amplitude: $V_{ramp} = V_S / k_{FF} = 60/15 = 4\ \text{V}$
- Error Amplifier Open-Loop DC Voltage Gain: $A_{OL} = 94\ \text{dB}\ (50118\ \text{V/V})$
- Gain Bandwidth Product: $GBW = 6.5\ \text{MHz}$
- Error Amplifier Open-Loop Compensation Pole: $f_{pe} = \dfrac{GBW}{A_{OL}} = \dfrac{6.5 \cdot 10^6}{50118} = 129.7\ \text{Hz}$

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
PWM CONTROL						
DC_{100kHz}	Maximum duty cycle	F_{SW} = 100 kHz, 6 V ≤ V_{VIN} ≤ 60 V	98	99		%
DC_{400kHz}		F_{SW} = 400 kHz, 6 V ≤ V_{VIN} ≤ 60 V	90	94		%
$V_{RAMP(min)}$	RAMP valley voltage (COMP at 0% duty cycle)			300		mV
k_{FF}	PWM feedforward gain (V_{IN} / V_{RAMP})	6 V ≤ V_{VIN} ≤ 100 V		15		V/V
ERROR AMPLIFIER						
V_{REF}	FB reference Voltage	FB connected to COMP	792	800	808	mV
AVOL	DC gain			94		dB
GBW	Unity gain bandwidth			6.5		MHz



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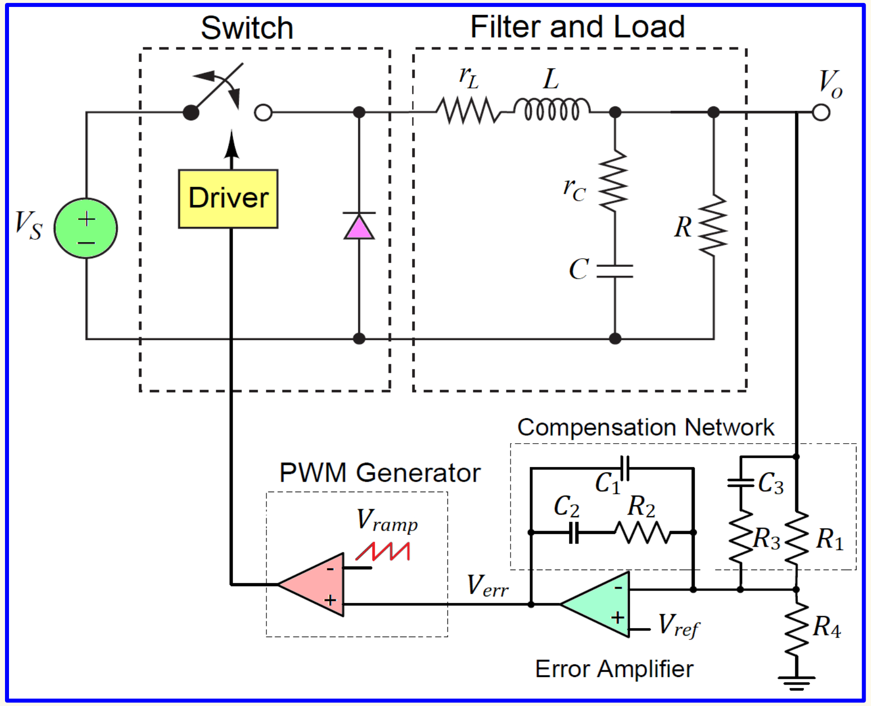
- Design for a **crossover frequency** of 10 kHz and a **phase margin** of $55°$

$V_{ref} = 0.8\text{ V}$

$\phi_{comp} = 201°$

$|G_3(j\omega_t)| = 1.438$



Solution: ❑ Calculations *Compensator Values – K Factor Method*

$$G_3(j\omega) = -\frac{1}{R_3 C_1} \cdot \frac{(j\omega + \omega_z)^2}{j\omega(j\omega + \omega_p)^2}$$

- **Step 1:** *Calculate the K Factor*

$$K = \tan^2\left(\frac{\phi_{comp} + 90°}{4}\right) = \tan^2\left(\frac{201° + 90°}{4}\right) = 10.4$$

- **Step 2:** *Calculate Resistor R_2* Let $R_1 = 200\text{ k}\Omega$

$$R_2 = \frac{|G_3(j\omega_t)| \cdot R_1}{\sqrt{K}} = \frac{1.438 \cdot 2 \cdot 10^5}{\sqrt{10.4}} = 89.18\text{ k}\Omega$$

- **Step 3:** *Calculate Capacitor C_1*

$$C_1 = \frac{1}{\omega_t R_2 \sqrt{K}} = \frac{1}{2\pi \cdot 10^4 \cdot 89.18 \cdot 10^3 \cdot \sqrt{10.4}} = 55.34\text{ pF}$$

- **Step 4:** *Calculate Capacitor C_2*

$$C_2 = \frac{\sqrt{K}}{\omega_t R_2} = \frac{\sqrt{10.4}}{2\pi \cdot 10^4 \cdot 89.18 \cdot 10^3} = 575.5\text{ pF}$$

- **Step 5:** *Calculate Capacitor C_3*

$$C_3 = \frac{\sqrt{K}}{\omega_t R_1} = \frac{\sqrt{10.4}}{2\pi \cdot 10^4 \cdot 2 \cdot 10^5} = 256.6\text{ pF}$$

- **Step 6:** *Calculate Resistor R_3*

$$R_3 = \frac{1}{\omega_t C_3 \sqrt{K}} = \frac{1}{2\pi \cdot 10^4 \cdot 256.6 \cdot 10^{-12} \cdot \sqrt{10.4}}$$

$$R_3 = 19.23\text{ k}\Omega$$

- **Step 7:** *Calculate Resistor R_4*

$$\frac{V_{ref}}{V_o} = \frac{R_4}{R_1 + R_4}$$

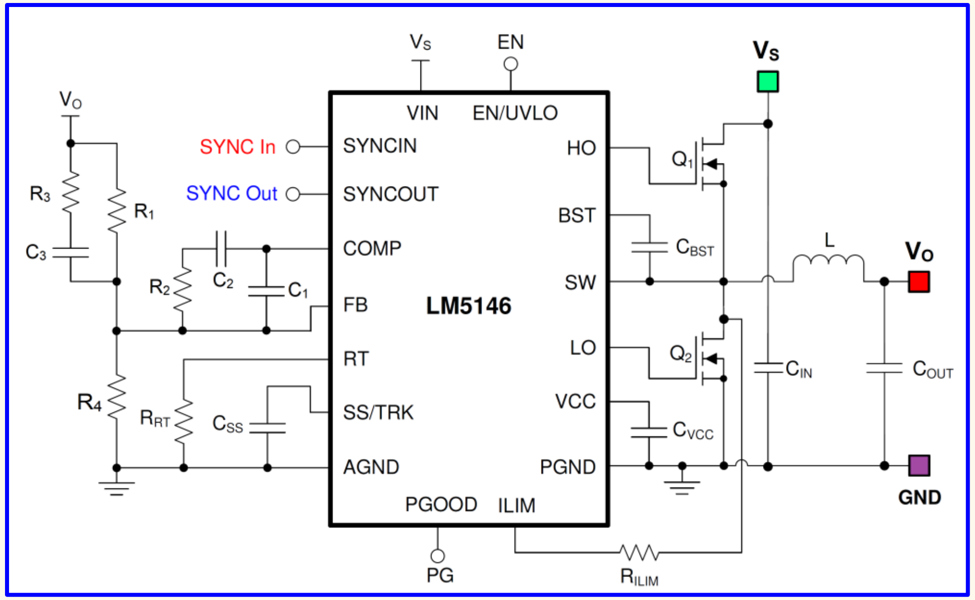
$$V_{ref}(R_1 + R_4) = V_o R_4$$

$$V_{ref} R_1 = (V_o - V_{ref}) R_4$$

$$R_4 = \frac{V_{ref}}{V_o - V_{ref}} R_1$$

$$R_4 = \frac{0.8}{15 - 0.8} \cdot 2 \cdot 10^5$$

$$R_4 = 11.27\text{ k}\Omega$$



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Simulations

MATLAB/Simulink

Given:

- DC input voltage $V_S = 60\ \mathrm{V}$
- Average output voltage $V_o = 15\ \mathrm{V}$
- Average output current $I_o = 2\ \mathrm{A}$
- Maximum percentage peak-peak output ripple voltage $\Delta V_o / V_o = 0.01\ (1\%)$

Buck Converter Component Values & Selections:

$L = 300\ \mu\mathrm{H}$ $\quad r_L = 25\ \mathrm{m\Omega}$ $\quad R = V_o/I_o = 7.5\ \Omega$

$C = 20\ \mu\mathrm{F}$ $\quad r_C = 400\ \mathrm{m\Omega}$ $\quad f_s = 100\ \mathrm{kHz}$ (selected)

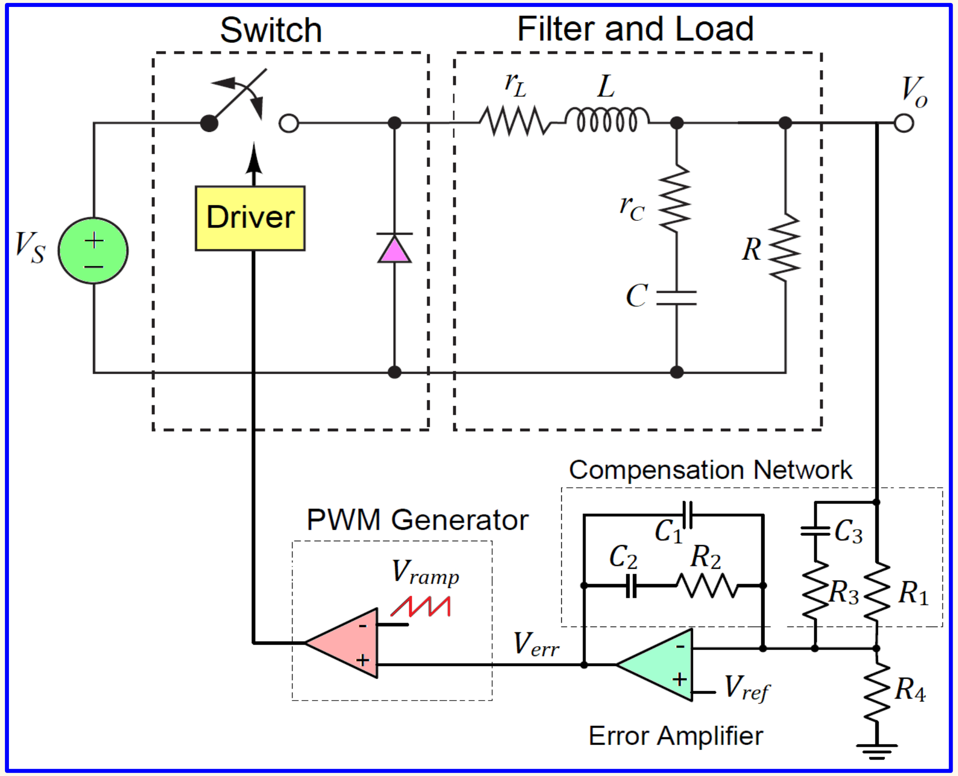
Design Objective:

- **Type 3 Compensated Error Amplifier** for a **stable control system**.
- Design for a **crossover frequency** of $10\ \mathrm{kHz}$ and a **phase margin** of $55°$

$V_{ref} = 0.8\ \mathrm{V}$

$\phi_{comp} = 201°$

$|G_3(j\omega_t)| = 1.438$



Solution: ❑ Simulations – MATLAB Script

```
s = tf('s'); % Define a Laplace parameter

%**SPECIFICATIONS**
Vs = 60; Vo = 15; % Input and output DC voltage values

%**SELECTIONS**
f_s = 1e5; % Switching frequency

%**BUCK CONTROLLER IC: LM5146**
Vref = 0.8; % Reference voltage from Buck Controller LM5146
Vramp = Vs/15; %Ramp voltage amplitude from Buck Controller LM5146
A_DC = 50119; % Error amplifier open-loop DC gain
GBW = 6.5e6; % Error amplifier gain bandwidth product
f_pe = GBW/A_DC; % Error amplifier open-loop pole: f_pe = 129.7 Hz.

%**COMPONENTS**
R = 7.5; L = 300e-6; C = 20e-6; % Buck converter output filter values
r_C = 400e-3; % ESR of the capacitor C
r_L = 25e-3; % ESR of the inductor L

%**TRANSFER FUNCTIONS**
M = Vs/Vramp; % PWM modulator transfer function
F = (1+r_C*C*s)/(1+r_L/R + (L/R+(r_C+r_L)*C+r_C*r_L*C/R)*s + (1+r_C/R)*L*C*s^2); % Filter and Load

%**FREQUENCIES and PARAMETERS**
f_t = 0.1*f_s; % Crossover frequency (selected value one decade below f_s)
phase_comp = 201*pi/180; % Required compensator phase.
K_comp = 1/(M*0.04636); % Required compensator gain at the crossover frequency.
K = (tan(0.25*(phase_comp + pi/2)))^2; % K factor value.

%**CONTROLLER COMPONENT VALUES**
R1 = 2e5; % Upper resistor for reference voltage divider (selected).
R2 = K_comp*R1/sqrt(K);
C1 = 1/(2*pi*f_t*R2*sqrt(K));
C2 = sqrt(K)/(2*pi*f_t*R2);
C3 = sqrt(K)/(2*pi*f_t*R1);
R3 = 1/(2*pi*f_t*C3*sqrt(K));
R4 = Vref*R1/(Vo-Vref); % Lower resistor for reference voltage divider.

%**COMPENSATOR TRANSFER FUNCTION**
G_OL = -A_DC/(1+s/(2*pi*f_pe)); % Error amplifier transfer function
beta = (R1*R3*C1*s*(s+(C1+C2)/(C1*C2*R2))*(s+1/(R3*C3)))/((R1+R3)*(s+1/(R2*C2))*(s+1/((R1+R3)*C3)));
G_comp = G_OL/(1+G_OL*beta); % Compensator transfer function
G_loop = M*F*G_comp;
```

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$R_3 = 19.23\ \mathrm{k\Omega}$

$R_4 = 11.27\ \mathrm{k\Omega}$

$C_1 = 55.34\ \mathrm{pF}$

$C_2 = 575.5\ \mathrm{pF}$

$C_3 = 256.6\ \mathrm{pF}$

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
PWM CONTROL						
DC_{100kHz}	Maximum duty cycle	F_{SW} = 100 kHz, 6 V ≤ V_{VIN} ≤ 60 V	98	99		%
DC_{400kHz}		F_{SW} = 400 kHz, 6 V ≤ V_{VIN} ≤ 60 V	90	94		%
$V_{RAMP(min)}$	RAMP valley voltage (COMP at 0% duty cycle)			300		mV
k_{FF}	PWM feedforward gain (V_{IN} / V_{RAMP})	6 V ≤ V_{VIN} ≤ 100 V		15		V/V
ERROR AMPLIFIER						
V_{REF}	FB reference Voltage	FB connected to COMP	792	800	808	mV
AVOL	DC gain			94		dB
GBW	Unity gain bandwidth			6.5		MHz

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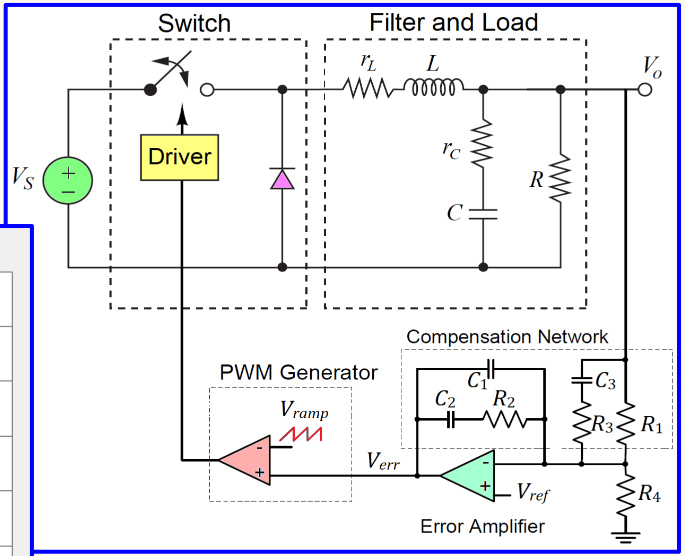
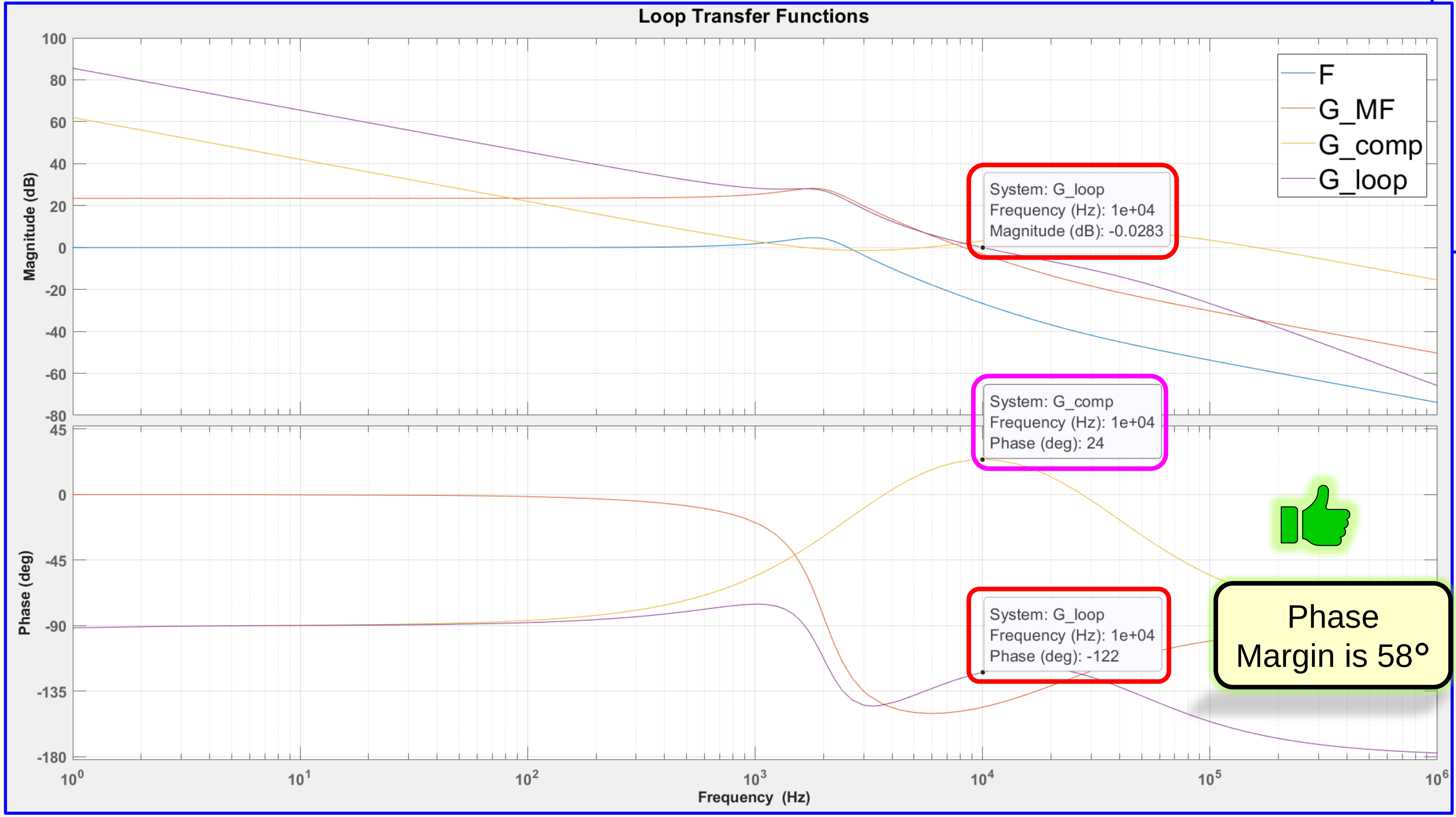


Solution: ❑ Simulations – Loop Transfer Functions



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Simulations

TINA-TI SPICE

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Design Objective:
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- Design for a **crossover frequency** of 10 kHz and a **phase margin** of 55°

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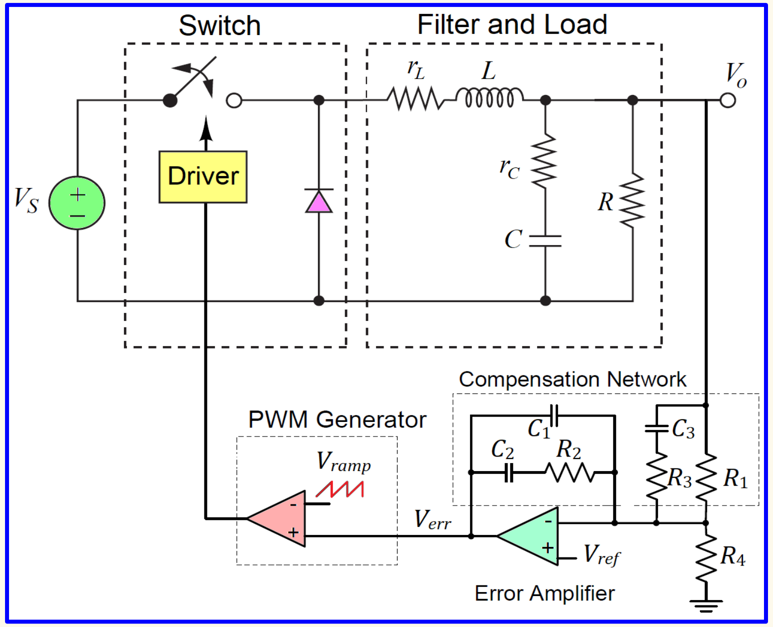
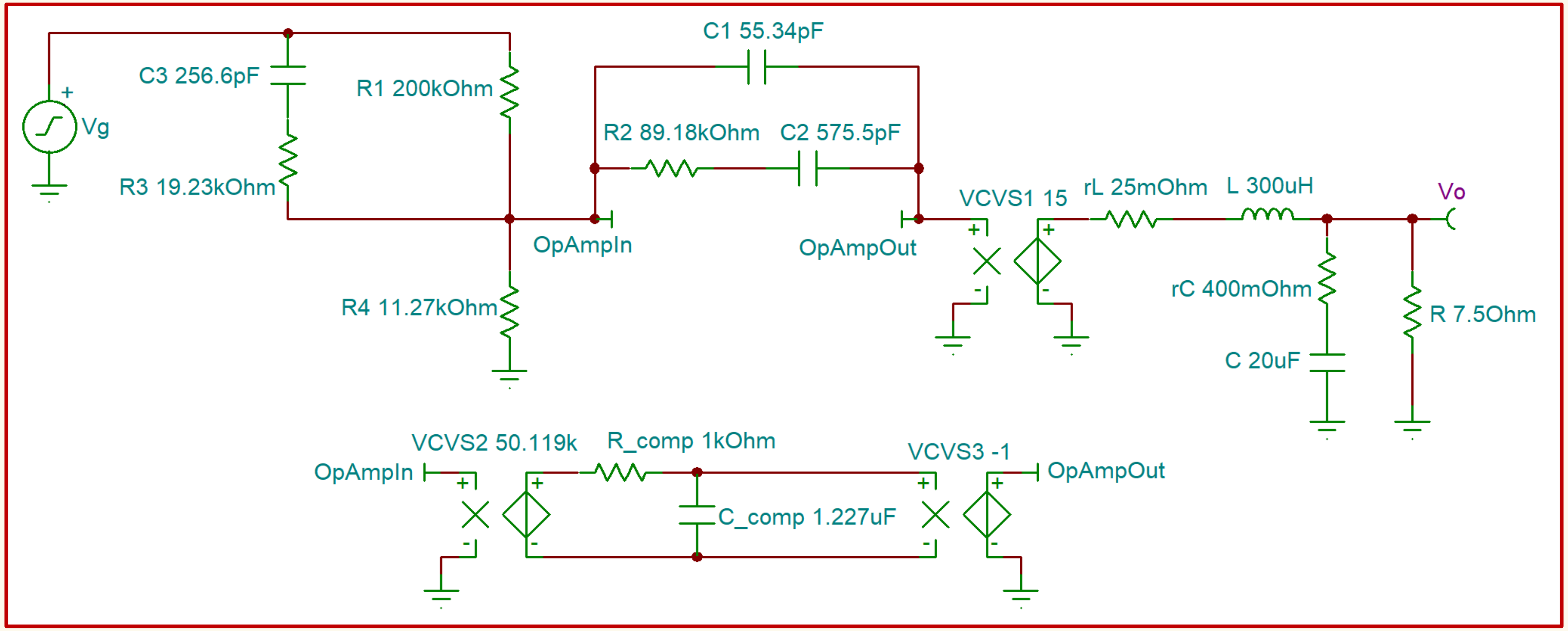


Solution: ❑ Simulations – Circuit



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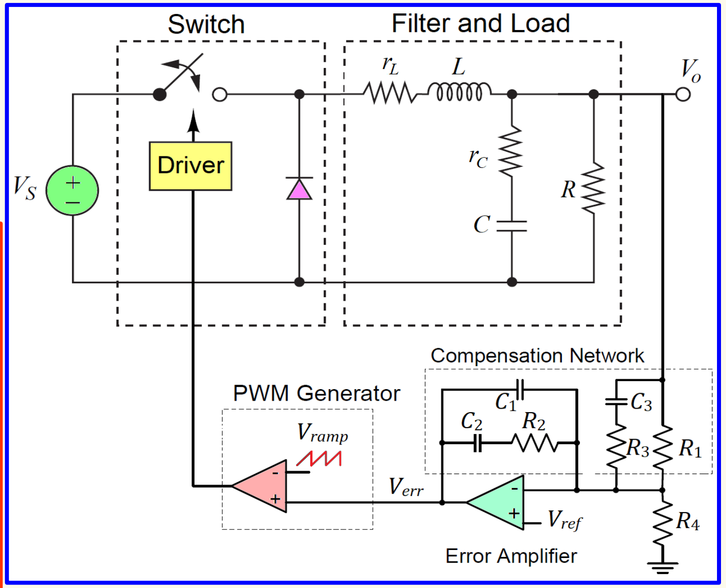

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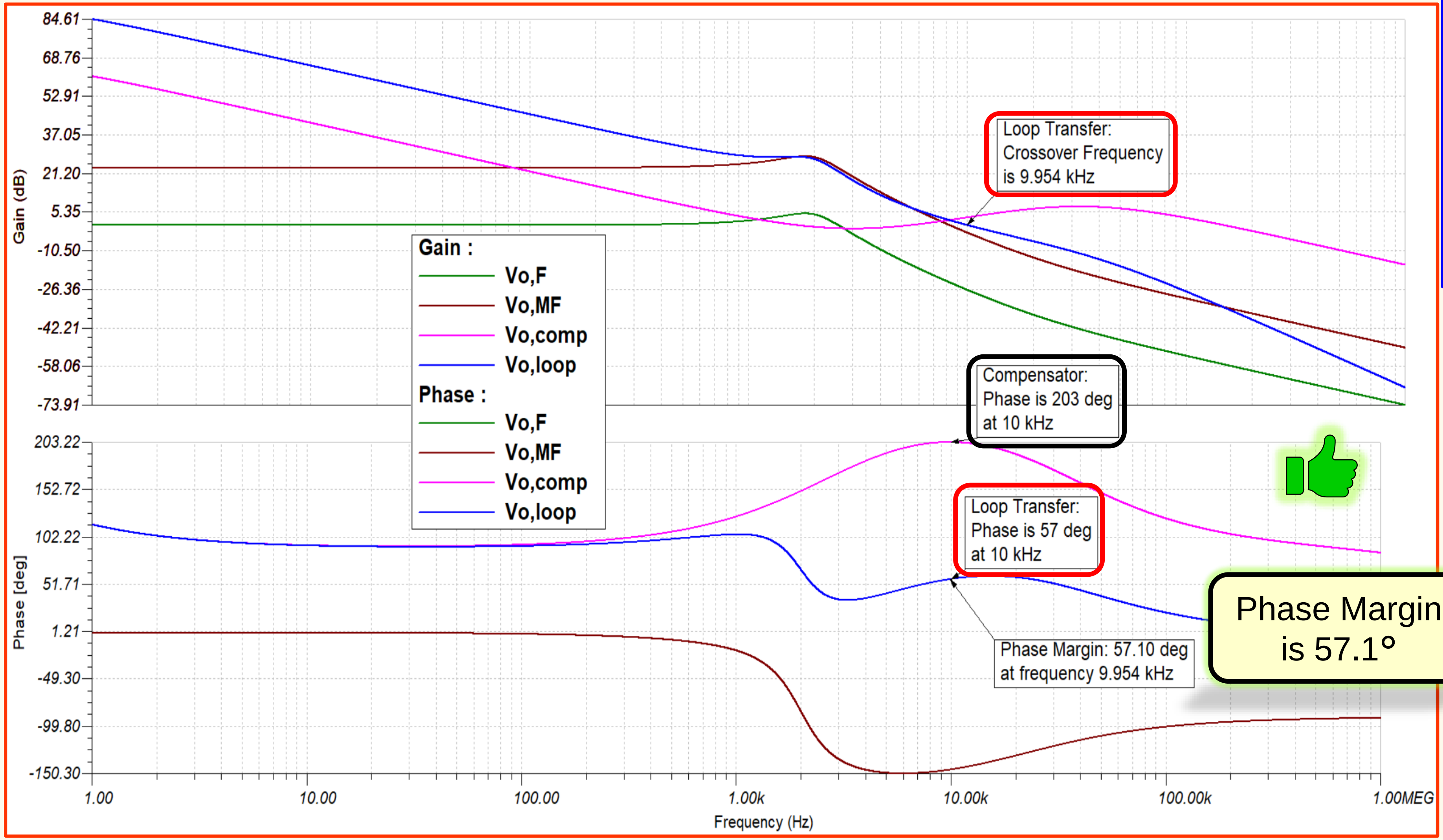

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